05-01 第2篇

阅读理解（说明文） Why Do Geese Fly in a V-Shape?

① On autumn and spring mornings in many parts of China, people looking up at the sky may see a strange, neat picture: a line of birds flying across the clouds in a tidy **V-shape**. They are usually **wild geese**, travelling thousands of kilometres between their summer homes in the north and their winter homes in the south. Why do they fly in such a clear pattern? Is it just beautiful, or does the V-shape actually help them in their long journey?

② Scientists have studied this question for more than fifty years. The first important clue came from the way birds' wings make air move. When a bird flaps its wings, the air on the **tips of its wings** is pushed upward in a swirl. In a V-shape, each bird behind the leader flies a little to the **side and just behind** the wings of the bird in front. There, the air is still spinning gently upward. The follower receives a small, **free lift**, the way a small boat is carried by a bigger boat's wake.

③ How much energy does this save? In 2014, a study in the journal *Nature* used tiny GPS recorders on **bald ibises** (a kind of bird that also flies in V-shape). The scientists found that birds in good V positions used **up to 20 percent less energy** than a single bird flying alone. Over a journey of a thousand kilometres, this can be the difference between arriving with strong muscles, or falling from the sky in the middle of the sea.

④ The V-shape also helps the birds **see one another**. Unlike cars on a straight road, birds do not have bumpers or mirrors. If they flew in a straight line, they could not watch their neighbours well. In a V, every bird can see the one in front and the one behind, **adjust speed** if somebody slows, and shout to the group if there is a storm or an enemy hawk below.

⑤ Interestingly, the leader of a V is not always the same bird. Flying at the front is the hardest work — the leader meets the strongest air with no free lift from anyone. So geese often **swap leaders** every few minutes. When the front bird is tired, she drops back; another bird, stronger now, moves to the front. This kind of **teamwork** is one of the reasons wild geese can cross oceans and mountain ranges.

⑥ Human engineers have noticed this clever trick. Some studies for future **airplanes** and for **military drones** suggest that flying in groups, with careful V spacing, could save fuel, exactly as it does for birds. The next time you see a line of geese cutting across a blue sky, do not only say, "What a pretty picture!" Remember that this V is, in fact, an **ancient flying machine** — built by nature long before the first human plane rose into the air.

根据文章内容，选择最佳答案。

1. What is the main topic of the article?

A. How to catch a wild goose.
B. Why wild geese fly in a V-shape and what scientists have found about it.
C. How to build a paper airplane.
D. Why birds sing in spring.

2. How does flying behind and to the side of another bird help a goose?

A. It makes the goose smaller.
B. The air there is spinning gently upward, giving the follower a small, free lift.
C. It helps the goose change colour.
D. It makes the goose eat more fish.

3. According to the 2014 study, about how much energy can birds in good V positions save compared with flying alone?

A. Up to 5%.
B. Up to 20%.
C. Up to 50%.
D. Up to 99%.

4. Why do geese often swap the leader in the V?

A. Because the leader is always the youngest.
B. Because the leader has the hardest work and gets tired, so others take turns.
C. Because the leader has broken wings.
D. Because the leader does not like rain.

5. What do human engineers learn from the V-shape?

A. Nothing at all.
B. That aeroplanes and drones flying in careful V groups could save fuel, just like geese.
C. That birds should be painted blue.
D. That people should stop building planes.

答案与解析

1.
【答案】B
【题型】主旨大意题
【定位】第①段末 + 全文
【解析】大雁为何排V字及科学发现。故选 B。

2.
【答案】B
【题型】细节理解题
【定位】第②段 "free lift"
【解析】获得向上气流的免费升力。故选 B。

3.
【答案】B
【题型】细节理解题
【定位】第③段 "20 percent less"
【解析】节省约20%能量。故选 B。

4.
【答案】B
【题型】细节理解题
【定位】第⑤段
【解析】领头太累，需要轮换。故选 B。

5.
【答案】B
【题型】细节理解题
【定位】第⑥段
【解析】飞机无人机编队可省燃油。故选 B。

词汇注释（超纲词）

- **goose** [n.] 大雁/鹅（复数 geese）
- **wake** [n.] 尾流
- **drone** [n.] 无人机